EI MegaProbe Language

a bioinformatics langauge

Isaac H. Lopez Diaz[1]

[1]Department of Computer Science
University of Puerto Rico

March 29, 2026

Agenda

- Introduction
- Methods
- Current Results
- Conclusions
- Future Work

Introduction

- El MegaProbe Language: Statically-typed, ML-like language with bioinformatics primitives
- Help The Blueprint of Life project with a language that can handle large genome sequences and process files

Methods

The language implementation follows the convention from Krishnamurthi [2].

- Lexing and parsing: parser combinators [1]
- Type Checking and Evaluating: Pattern Matching
- Dataframe: Apache Arrow API

Syntax

Operational Semantics

Current Results

- A (sort of) compiler
- DNA and RNA sequences as primitives, and higher-order functions like transcribe DNA sequence and reverse complement.
- (minimal) Apache Arrow bindings for fast reading and writing csv files, and producing Apache Arrow Tables (dataframes).

Conclusions

Future Work

- Add ADTs and pattern matching
- Necessary bindings to Apache Arrow
- Allow user to compile to OpenCL and implement
- Allow user to compile to NVIDIA and implement

Graham Hutton and Erik Meijer.
Monadic parser combinators, 2011.

Shriram Krishnamurthi.
Programming languages: Application and interpretation.
Brown University, 2017.